

Mehmet Güzel

SOFTWARE ENGINEER • BACKEND & DEVOPS

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🗣️ Languages: English (B2) | Turkish (native)

■ Summary

Software Engineer, class of 2026, focused on backend, distributed systems, and DevOps. I build backend services and REST APIs with Java/Spring Boot and Python, and work with relational and NoSQL databases, messaging systems, and cloud environments. I have built microservices and event-driven architectures. I write CI/CD pipelines with Docker and GitHub Actions and deploy applications to the cloud. I have experience with LLM integration and AI-powered product development. I care about writing secure, maintainable, and clean code.

■ Experience

Backend Developer Intern

August 2026 – September 2026

Codexist

Istanbul, Türkiye

- Researched GEO (Generative Engine Optimization) and designed the architecture of a system that measures how visible companies are in LLM responses relative to their competitors; prompts are sent repeatedly to multiple LLM providers, and results are evaluated with a statistical margin of error.
- Designed and built an asynchronous task architecture based on **Celery** and **Redis** for daily automated crawling and analysis jobs; developed workers and workflows that use a **sliding window** approach to reduce the margin of error in measurements obtained from repeated LLM queries.
- Turned the analysis pipeline into backend services with **Python** and **FastAPI** and integrated it with **PostgreSQL**; connected the resulting APIs to a **TypeScript/React** frontend and containerized the application with **Docker Compose**.

Application Security Long-Term Intern

June 2024 – July 2025

Privia Security

Istanbul, Türkiye

- Analyzed production web applications at the code and API level to identify security vulnerabilities, and took part in web application penetration testing engagements.
- Developed internal tools in **Python** to speed up the team's repetitive security testing processes, contributing to the automation of manual checks.
- Discovered a stored XSS vulnerability in a commercial product and reported it through responsible disclosure; the finding was officially assigned **CVE-2025-1301**.
- Worked with development teams to remediate the identified vulnerabilities, proposing actionable code-level fixes and contributing to secure development practices.

■ Projects

STOP — Event-Driven Microservice Platform (*B.Sc. graduation project*)

🐙 mehmettguzell/STOP

Java 21 · Spring Boot · Apache Kafka · PostgreSQL · Redis · Docker · GitHub Actions · AWS

- Built the backend of a football match organization platform as a microservice architecture, split into 5 independent **Spring Boot** services: **API Gateway**, **Identity**, **Match**, **Communication**, and **Notification**. Each service has its own **PostgreSQL** database, and schema changes are managed with **Flyway**.
- Implemented inter-service communication largely through **Kafka** events (more than 20 topics covering matches, participation, waiting lists, and chat); notifications are created when the Notification Service consumes these events.
- Each service verifies JWTs with its own RSA public key; the API Gateway provides Redis-backed rate limiting with **Bucket4j** and a **Resilience4j** circuit breaker.
- Developed match management (join requests, invitations, and a FIFO waiting list that automatically admits the next player when a spot opens), live chat over **WebSocket (STOMP)**, and a goalkeeper-call feature.
- Set up a **GitHub Actions** CI/CD pipeline that builds only the changed services and pushes them to **GHCR** tagged with the Git SHA; **release-please** generates versions and changelogs. I deploy to **AWS EC2** without SSH using **SSM Run Command**; secrets are fetched from **Secrets Manager** through an IAM instance role, with no static credentials in the repository or CI. A failed health check triggers an automatic rollback; services have per-service resource limits and **Prometheus/Actuator** metrics.

DevOps Todo App — End-to-End CI/CD Reference Pipeline

🐙 mehmettguzell/Devops-Todo-App

GitHub Actions · Docker · GHCR · AWS EC2 · IAM · SSM · OIDC · Python · TypeScript

- Built a **GitHub Actions** CI/CD pipeline covering testing, release, build, and deployment for a **Python/FastAPI**

backend and a TypeScript frontend: with **dorny/paths-filter** and a dynamic build matrix, only the changed services are tested and built. Wrote **multi-stage Dockerfiles** for the backend and frontend; images are pushed to **GHCR** with Git SHA and semver tags.

- Deploy to **AWS EC2** without SSH using **SSM Run Command**. A temporary IAM role is obtained via **OIDC**, so CI holds no static AWS keys. Production **.env** values are stored in **SSM Parameter Store** and fetched with the server's own IAM role.
- The application starts with **Docker Compose** using health checks, startup ordering, and resource limits. The deployment waits until services are healthy and fails the pipeline on error. I also wrote the bootstrap script that prepares the EC2 instance with Docker, the AWS CLI, and the SSM Agent.
- **release-please** generates versions and changelogs from Conventional Commits. **pytest**, **ruff**, and frontend lint run on pull requests.

Event-Driven Order Management Platform — Distributed Transactions [Event-Driven-Order-Management-Platform](#)

[Python](#) · [Django REST Framework](#) · [RabbitMQ](#) · [PostgreSQL](#) · [JWT](#) · [Nginx](#) · [Docker Compose](#)

- Built a commerce backend with 5 services (user, product, order, inventory, and payment); used **RabbitMQ** for asynchronous stock and payment processing, the **Saga pattern** with compensating transactions for consistency, and row-level locking for concurrent stock updates.
- Provided service routing, rate limiting, and JWT authentication through an **Nginx API Gateway**; services are isolated with a database-per-service design.

Inventory Management Microservices — Polyglot Persistence & API Gateway [mehmettguzell/Microservice_Project](#)

[Java](#) · [Spring Boot](#) · [Spring Cloud Gateway](#) · [MongoDB](#) · [MySQL](#) · [Flyway](#) · [Testcontainers](#) · [OpenAPI](#) · [Next.js](#)

- Built a Spring Boot system with 4 services (gateway, product, inventory, and order); used **MongoDB** for products and **MySQL** for inventory and order (polyglot persistence), and centralized routing, CORS, and OAuth2 in **Spring Cloud Gateway**.
- Wrote integration tests with **Testcontainers** + **RestAssured** against real MySQL and MongoDB instances; prepared OpenAPI documentation and a **Next.js** + **TailwindCSS** frontend that works through the gateway.

■ Technical Skills

Languages: Java 21, Python, Go, TypeScript, SQL, Bash

Backend: Spring Boot, Spring Cloud Gateway, Spring Security, Spring Data JPA, FastAPI, Django REST Framework, REST API, OpenAPI, Microservices, Event-Driven Architecture, Saga pattern, Resilience4j, Bucket4j

Security: OWASP Top 10, secure code review, web application pentest, JWT (RSA), OAuth2/OIDC, rate limiting, secret management (Secrets Manager, SSM), responsible disclosure (CVE-2025-1301)

Messaging & Data: Apache Kafka, RabbitMQ, Celery, PostgreSQL, MySQL, MongoDB, Redis, Flyway

DevOps & CI/CD: GitHub Actions, Docker, Docker Compose, GHCR, release-please, Nginx, Linux (Ubuntu), Prometheus

Cloud: AWS EC2, IAM, SSM, Secrets Manager, OIDC

Frontend: React, React Native, Next.js, TailwindCSS

Testing: JUnit, Testcontainers, RestAssured, pytest

■ Education

Maltepe University

B.Sc. in Software Engineering (English) — GPA: 3.11/4.00

September 2021 – June 2026

Istanbul, Türkiye

■ Achievements & Certifications

Competitions: **Future Commerce AIdeathon** — organized by the Republic of Türkiye Ministry of Trade as part of Türkiye E-Commerce Week; worked on an AI-based stock optimization solution for the dead stock problem

Certifications: CSD Java Programming Course · BTK Akademi Cybersecurity Camp

■ References

İlhan Zeki Köroğlu: Group CFO — Başoğlu Kablo ve Profil San. ve Tic. A.Ş.

Ensar Gül: Professor — Beykoz University

Volkan Günaydın: Network Corporate Services Management Manager — Türk Telekom

Alper Bekar: Co-Founder — Codexist

Ömer Kepenek: Senior Security Engineer — Garanti BBVA Technology

Zeynel Abidin Fincan: Founder — Upsoft Solutions

Barış Gökhan: Portfolio Manager — Experian